

Reinforcement Learning for GUI Testing

From Grid Worlds to Web Automation

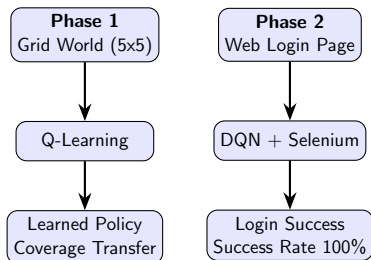
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May 2, 2026

Project Overview



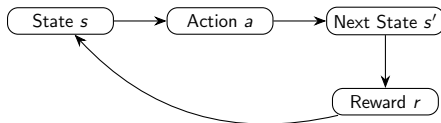
Grid World Environment

S	B	M	F	X
-	-	-	-	X
-	-	C	-	X
-	-	-	-	G
X	X	X	X	E

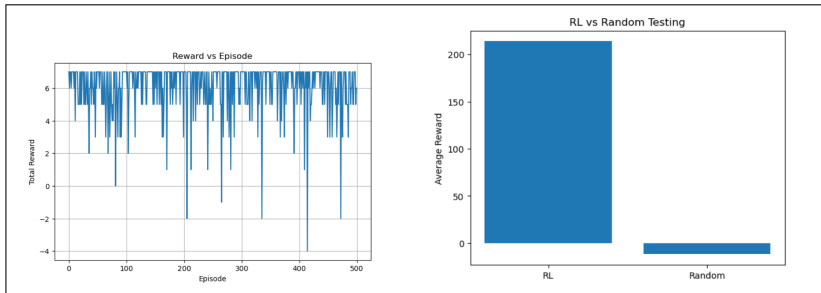
Legend: S=Start, B=Button, M=Menu, F=Form,
C=Crash, G=Goal, E=Exit, X=Blocked

Q-Learning Update

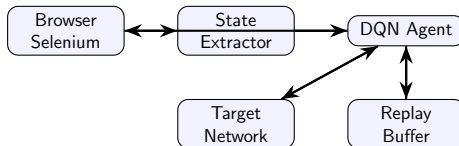
$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$



Grid World Results



Web Automation Architecture



Web Login Page

Login Page

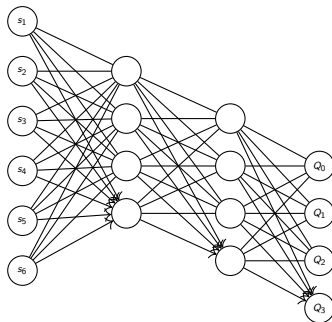
Username: _____

Password: _____

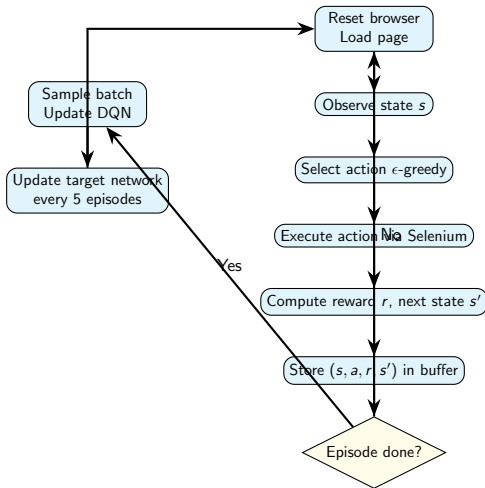
Login

Correct credentials: admin / 1234

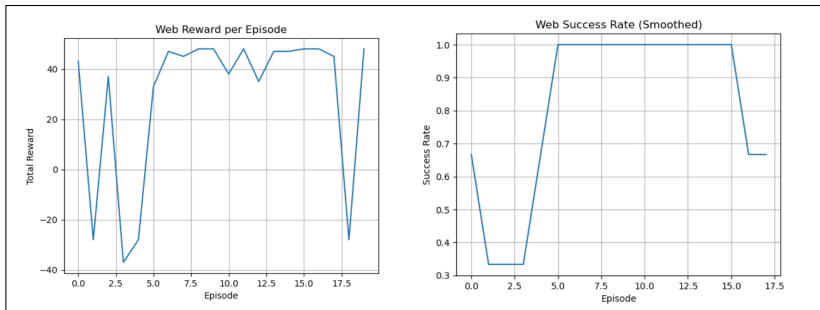
DQN Architecture



Training Loop (Web DQN)



Web Automation Results



Metric	Final Value
Success rate (last 20 episodes)	1.00
Average reward (last 10 episodes)	104
Episodes to first success	2

Conclusion

Grid World
Q-learning learns optimal policy

Web DQN learns to log in automatically

Transfer learning works across similar layouts

Future: more complex UIs, pop-ups, dynamic content